Learning Experience Questionnaire

Simulation-Based Instruction Version

Participant ID: _____

Section A. Perceived Learning Support (PLS)

PLS1. The simulation-based teaching system helped me better understand the mechanisms of traffic conflict formation in ramp merging scenarios.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

PLS2. The system improved my understanding of vehicle interactions in weaving sections.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

PLS3. The simulation-based activities helped me connect theoretical knowledge with real-world traffic scenarios.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

PLS4. Compared with traditional classroom teaching, the system made complex traffic processes easier to understand.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

PLS5. The post-simulation debriefing helped me more clearly identify the causes of traffic conflicts.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

Section B. Cognitive Engagement (CE)

CE1. Compared with traditional lectures, I was more focused during the simulation-based learning process.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

CE2. The simulation tasks encouraged me to actively think about how traffic conflicts are formed.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

CE3. I was more willing to participate in discussions and analysis during the simulation-based class.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

CE4. The simulation environment increased my interest in learning traffic conflict-related content.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

CE5. In the simulation scenarios, I was willing to invest more effort in understanding vehicle interactions.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

Section C. Perceived Realism and Interactivity (PRI)

PRI1. The ramp merging and weaving scenarios in the simulation system were realistic and supported my learning.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

PRI2. The dynamic interactions between vehicles in the scenarios helped me understand how traffic conflicts are formed.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

PRI3. Compared with static diagrams or instructional videos, the simulation tasks provided a stronger sense of involvement.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

PRI4. The system allowed me to observe traffic processes that are difficult to understand in traditional teaching.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

PRI5. The interactive experience made the learning process more intuitive.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

Section D. Learning Satisfaction and Behavioral Intention (LSBI)

LSBI1. I am satisfied with the use of the simulation-based teaching system in this course.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

LSBI2. I believe this teaching approach has practical value for transportation engineering courses.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

LSBI3. I would like to have similar simulation-based learning activities in other transportation-related courses.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

LSBI4. The system improved my overall learning experience in this topic.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree

LSBI5. I would recommend this teaching approach to other students.

[ ] Strongly disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly agree